

RailSync – Train Reservation & Dynamic Seat Management System

Programming in Java (CSE1007) Project Report

1. Project Information

- **Project Title:** RailSync – Train Reservation & Dynamic Seat Management System
 - **Course Name:** Programming in Java
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 - **Institution:** Vellore Institute of Technology (VIT Bhopal University)
 - **Academic Year:** 2025–2026
 - **Technology Stack:** Core Java (JDK 17+), Java Swing (`javax.swing` , `java.awt`), Java Serialization
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2. Introduction

Railway passenger ticketing systems are essential computerized applications that handle high daily volumes of passengers, manage limited seating across various coach types, maintain waiting queues, and process ticket cancellations.

RailSync is a desktop application developed in Core Java that simulates train reservation and seat management operations. It models the end-to-end ticketing process:

- Searching trains and schedules between railway stations.
 - Booking tickets for individual and multiple passengers with berth preferences.
 - Allocating confirmed coach berths or placing passengers into **Reservation Against Cancellation (RAC)** and **Waiting List (WL)** queues.
 - Calculating distance-based fares with statutory age-based concessions.
 - Processing cancellations, generating refund receipts, and automatically promoting RAC and Waiting List passengers.
 - Preventing duplicate seat assignments during simultaneous bookings using thread synchronization.
 - Providing dual user interfaces: an interactive command-line interface (CLI) and a graphical user interface (GUI) built with Java Swing.
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3. Problem Statement

Most introductory student projects treat railway booking as a simple counter decrement. However, a realistic railway reservation system must handle several real-world constraints:

1. **Multi-tier Seat Allocation:** When physical coach seats are filled, reservation requests must transition to shared RAC seats and subsequent Waiting List queues rather than being outright rejected.
 2. **Cancellation & Queue Cascade:** When a confirmed passenger cancels their booking, the vacant berth must not be wasted. The system must automatically promote the first RAC passenger to confirmed status, and the first Waiting List passenger to RAC.
 3. **Race Conditions & Concurrency:** When multiple users attempt to book the last available berth at the same time, simultaneous threads can assign the same seat to two passengers unless access to the train inventory is synchronized.
 4. **Dynamic Fare & Concession Calculation:** Fares vary based on distance, train type multipliers, and passenger demographics (e.g. 40% senior citizen concession, free travel for infants).
 5. **State Persistence:** Train fleet information, station networks, and booking records must be saved to disk so data is retained across application restarts.
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4. Project Objectives

- Design and implement a train reservation simulation using standard Core Java libraries.
- Demonstrate **Object-Oriented Programming (OOP)** principles: abstraction, inheritance, encapsulation, and polymorphism.
- Implement data structures from the **Java Collections Framework** (`HashMap` , `LinkedList` , `ArrayList` , `HashSet`).
- Apply **multithreading and synchronization** to ensure thread-safe seat allocation.
- Implement robust **exception handling** using custom checked exception classes.
- Persist system state using **Java Object Serialization** (`ObjectInputStream` , `ObjectOutputStream`).
- Provide both a terminal-based CLI for fast operations and a Java Swing GUI for graphical user interaction.

5. Technologies & Java Concepts Used

Concept / Technology	Implementation in RailSync
Inheritance & Abstraction	Abstract base class <code>Train</code> extended by concrete train types: <code>RajdhaniExpress</code> , <code>ShatabdiExpress</code> , <code>VandeBharatExpress</code> , <code>SuperfastExpress</code> , and <code>ExpressTrain</code> .
Polymorphism & Interfaces	<code>FareCalculator</code> interface implemented by <code>DynamicFareCalculator</code> . Train subclasses override speed, base rates, and catering charges.
Encapsulation & Validation	Private model fields with getters/setters, defensive copying, and input validation in <code>ValidationUtils</code> .
Collections Framework	<code>HashMap</code> for $O(1)$ PNR and station lookups; <code>LinkedList</code> as FIFO queues for RAC and Waiting List; <code>ArrayList</code> for seat and train rosters.
Multithreading & Synchronization	<code>Thread</code> , <code>Runnable</code> , and fine-grained <code>synchronized (train)</code> blocks in <code>ReservationManager</code> preventing race conditions.
Custom Exception Handling	Custom checked exceptions under <code>RailSyncException</code> (<code>SeatUnavailableException</code> , <code>InvalidPNRException</code> , <code>CancellationNotAllowedException</code>).
File I/O & Serialization	Object serialization to <code>data/railsync_data.ser</code> for persistence; formatted reservation slip and report export via <code>FileWriter</code> .
GUI Development	Java Swing components (<code>JFrame</code> , <code>JPanel</code> , <code>JTabbedPane</code> , <code>JTable</code> , <code>CardLayout</code>) with clean layouts.

6. System Design & Flowcharts

6.1 Basic System Architecture

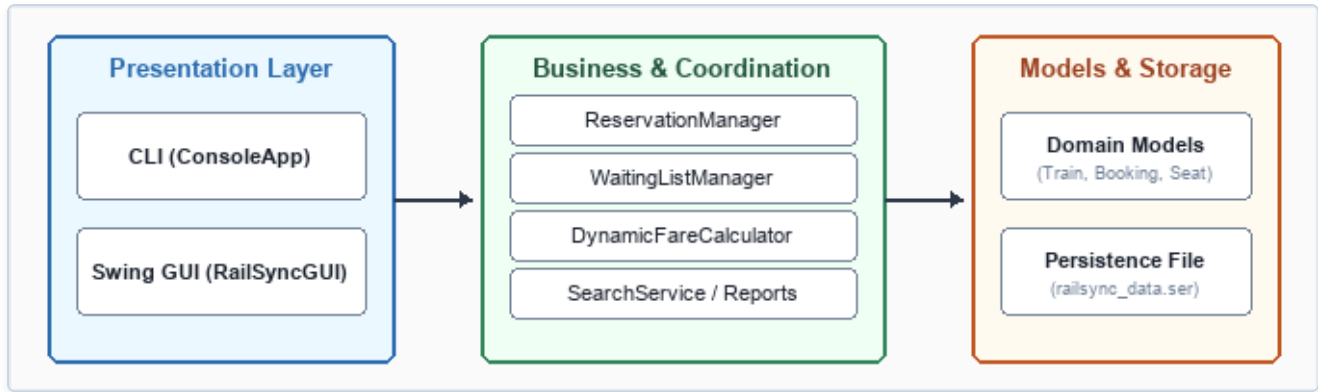


Figure 6.1: RailSync Layered System Architecture



Figure 6.1: RailSync Layered System Architecture

6.2 Train Search Flowchart

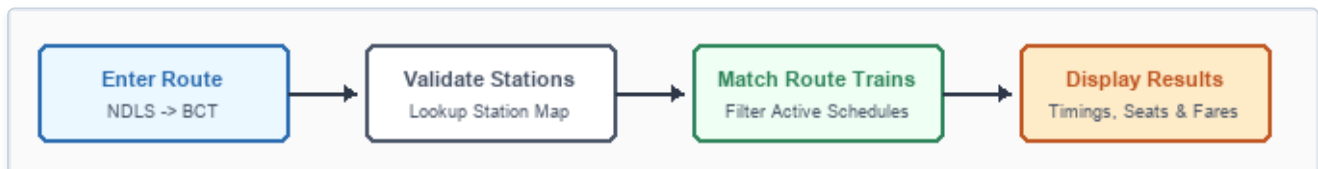


Figure 6.2: Train Route Search Workflow

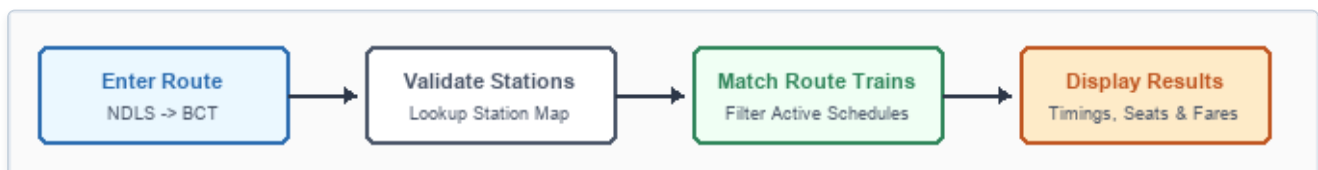


Figure 6.2: Train Route Search Workflow

6.3 Ticket Booking Flowchart

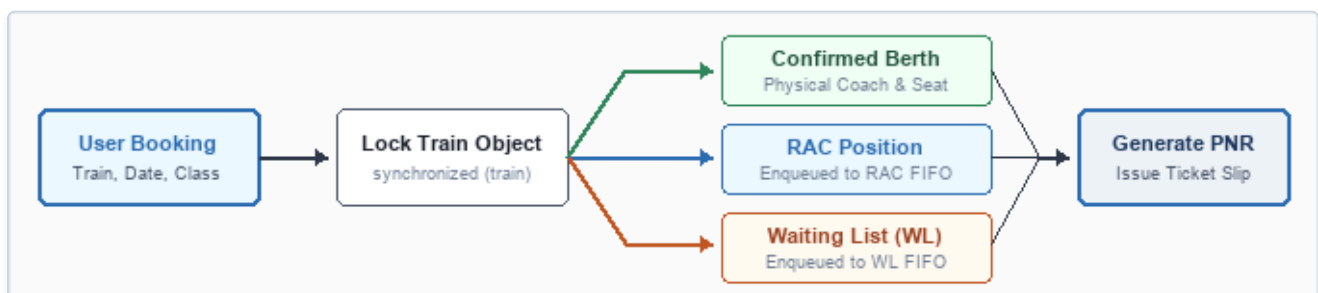


Figure 6.3: Ticket Booking & Tiered Allocation Flowchart

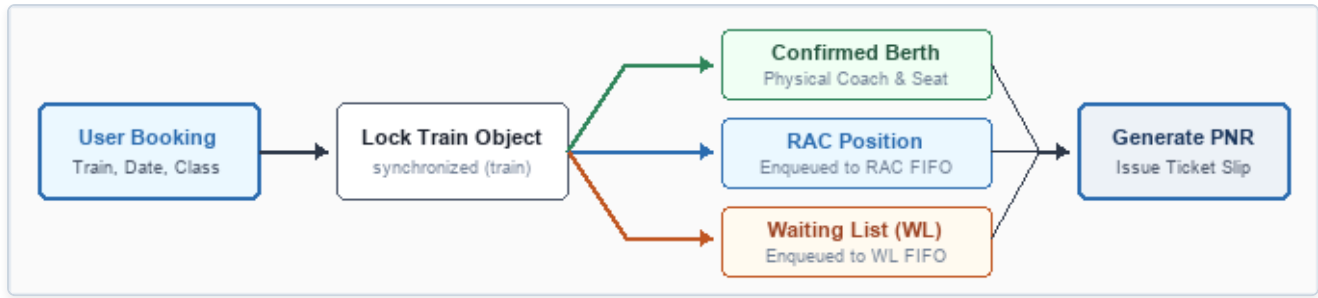


Figure 6.3: Ticket Booking & Tiered Allocation Flowchart

6.4 Ticket Cancellation & Promotion Cascade Flowchart



Figure 6.4: Cancellation & Queue Cascade Workflow



Figure 6.4: Cancellation & Queue Cascade Workflow

7. Main Modules

- 1. Train Search & Scheduling Module:** Enables route lookups between stations with departure/arrival timings, journey duration, and seat availability.
- 2. Booking & Berth Allocation Module:** Processes multi-passenger bookings, assigns coach berth numbers (Lower, Middle, Upper, Side Lower, Side Upper, Window, Aisle), and generates unique PNR numbers.
- 3. Queue Management Module:** Implements FIFO queues for RAC and Waiting List passengers when confirmed berths are fully booked.
- 4. Dynamic Fare & Concession Engine:** Computes distance-based fares with travel class multipliers, catering fees, and age-based statutory concessions (Senior Citizen 40%, Child 50%, Infant free).
- 5. Cancellation & Refund Module:** Computes time-tiered cancellation deductions (10% to 50%) and triggers automatic two-tier queue promotions.
- 6. Persistence & Export Module:** Serializes system state to disk and formats electronic reservation slips (tickets) and summary reports.
- 7. Concurrency Simulation Module:** Spawns concurrent worker threads booking seats simultaneously to demonstrate that `synchronized` blocks prevent double-booking.

8. Implementation Details & OOP Concepts

8.1 Abstraction and Inheritance

The application defines an abstract class `Train` containing common properties: train number, train name, source, destination, departure time, arrival time, distance, and coach roster. Specialized train categories inherit from `Train`:

- `RajdhaniExpress`: High-speed overnight train with dynamic catering surcharges and premium 1A, 2A, and 3A coaches.

- `ShatabdiExpress` : Daytime express with Chair Car (CC) and Executive 1A seating.
- `VandeBharatExpress` : Semi-high-speed modern train with catering and Chair Car accommodation.
- `SuperfastExpress` & `ExpressTrain` : Intercity long-distance trains featuring Sleeper (SL) and AC coaches.

8.2 Polymorphism

Each subclass overrides methods such as `getBaseFareRatePerKm()` and `calculateCateringCharge()` to supply specialized pricing rules. At runtime, the fare calculator polymorphically evaluates the exact subclass behavior without type-casting.

8.3 Thread Safety via Monitor Locks

To prevent race conditions during simultaneous bookings on the same train, seat allocation logic is enclosed inside a fine-grained monitor lock:

```
java
synchronized (train) {
// 1. Search available coach seats
// 2. Allocate berth or transition to RAC / Waiting List
// 3. Mark seat as booked
}
```

Synchronizing per-train rather than locking the entire reservation manager allows concurrent bookings on different trains to proceed in parallel without blocking each other.

9. Testing & Verification

The project includes an automated test runner (`TestRunner.java`) that executes 7 test scenarios validating all critical business rules:

Test #	Test Case Description	Verified Logic	Result
Test 1	Ticket Booking & Seat Allocation	Confirmed booking assigns non-null PNR, coach ID, and berth number.	PASS
Test 2	RAC Allocation upon Full Capacity	When physical seats are full, next booking correctly receives <code>RAC</code> status.	PASS
Test 3	Waiting List Allocation & Limit Check	When RAC is full, booking enters <code>WAITING_LIST</code> ; when WL is full, request is rejected.	PASS
Test 4	Cancellation & Cascade Promotion	Cancelling a confirmed ticket promotes RAC to Confirmed, and WL to RAC.	PASS
Test 5	Fare & Concession Calculation	Correct concession applied for Senior Citizens (40%) and Infants (100%).	PASS
Test 6	Multithreaded Booking Safety	10 concurrent threads compete for 4 seats; zero duplicate seat assignments occur.	PASS
Test 7	System State Persistence	State is successfully serialized to file, restored to memory, and integrity verified.	PASS

Verification Summary: 7 Passed, 0 Failed.

10. Application Screenshots & Results

The following screenshots are captured from the actual running RailSync application.

10.1 Main Menu & Application Start

The interactive terminal application displays the main menu with options for train search, catalog inspection, booking, PNR enquiries, cancellations, queue inspection, and fare calculation.

```
Command Prompt - RailSync Application Start & Main Menu

=====
RailSync - Interactive Command-Line Console Application
Core Java Train Reservation System
=====
----- MAIN MENU -----
1. Search Trains (by Source & Destination)
2. View All Trains Catalog
3. Book Train Ticket
4. PNR Status Enquiry
5. Cancel Ticket (with Promotion Cascade & Refund)
6. View RAC & Waiting List Queue Status
7. Dynamic Fare & Concession Calculator
8. Save System State to Disk
9. Exit
-----
Enter your choice (1-9):
Thank you for using RailSync. Goodbye!
```

Main Menu

10.2 Train Route Search

Demonstrates searching for direct trains between New Delhi (**NDLS**) and Mumbai Central (**BCT**), displaying train numbers, departure/arrival schedules, and available seat classes.

```
Command Prompt - RailSync Train Search (NDLS -> BCT)

Enter your choice (1-9): 1
--- SEARCH TRAINS ---
Enter Source Station Code (e.g. NDLS, BCT, HWH): Enter Destination Station Code (e.g. SBC, BSB, NDLS):
Found 1 train(s) on route NDLS -> BCT:
Train No | Name | Type | Departs | Arrives | Classes
-----
12952 | Mumbai Rajdhani | Rajdhani Express | 16:55 | 08:35 | 1A 2A 3A
```

Train Search

10.3 Train Fleet Catalog

Displays the complete registered train roster showing routes, train categories, and journey distances.

```
Command Prompt - RailSync Train Catalog

Enter your choice (1-9): 2
--- TRAIN CATALOG (9 Trains Registered) ---
Train No | Name | Type | Source | Destination | Distance
-----
12628 | Karnataka Express | Superfast Express | NDLS (New Delhi) | SBC (KSR Bengaluru) | 2400.0 km
12933 | Karnavati Express | Mail/Express | BCT (Mumbai Central) | ADI (Ahmedabad Junction) | 492.0 km
12626 | Kerala Express | Superfast Express | NDLS (New Delhi) | TVC (Thiruvananthapuram Central) | 3050.0 km
12002 | Bhopal Shatabdi | Shatabdi Express | NDLS (New Delhi) | BPL (Bhopal Junction) | 705.0 km
12301 | Howrah Rajdhani | Rajdhani Express | HWH (Howrah Junction) | NDLS (New Delhi) | 1447.0 km
12137 | Punjab Mail | Mail/Express | BCT (Mumbai Central) | NDLS (New Delhi) | 1540.0 km
12951 | NDLS Rajdhani Express | Rajdhani Express | BCT (Mumbai Central) | NDLS (New Delhi) | 1384.0 km
12952 | Mumbai Rajdhani | Rajdhani Express | NDLS (New Delhi) | BCT (Mumbai Central) | 1384.0 km
22436 | Vande Bharat Express | Vande Bharat Express | NDLS (New Delhi) | BSB (Varanasi Junction) | 780.0 km
```

Train Catalog

10.4 Ticket Booking & Seat Allocation

Shows booking a confirmed ticket on the Mumbai Rajdhani (12952) in AC First Class (1A) with passenger details, Lower Berth allocation (H1-1), unique PNR generation, and total fare.

```
Command Prompt - RailSync Ticket Booking & Confirmation

Enter your choice (1-9): 3
--- BOOK TICKET ---
Enter Train Number (e.g. 12952, 22436, 12628): Selected Train: 12952 - Mumbai Rajdhani (NDLS -> BCT)
Enter Journey Date (YYYY-MM-DD, press Enter for 2026-09-18):
Available Classes on this train:
  1. 1A - AC First Class (1A) (Confirmed seats: 18, RAC cap: 4)
  2. 2A - AC 2 Tier (2A) (Confirmed seats: 70, RAC cap: 12)
  3. 3A - AC 3 Tier (3A) (Confirmed seats: 96, RAC cap: 16)
Select Class (1-3): Enter number of passengers (1-6):
Passenger #1 Details:
  Full Name: Age (0-125): Gender (M/F/O): Mobile Number (10 digits): Berth Preference: 1) Lower 2) Middle 3) Upper 4) Side Lower 5) Side Upper 6) Window 7) Aisle
  Choice (1-7, default 1):
Processing reservation transaction...

===== BOOKING CONFIRMATION =====
PNR Number      : RS691564
Booking ID      : BK10004
Train           : 12952 - Mumbai Rajdhani
Route           : NDLS -> BCT
Journey Date    : 2026-09-18
Travel Class    : AC First Class (1A)
Total Amount Paid : ₹4880.00

Allocated Passenger Tickets:
Ticket ID | Passenger | Age | Status | Seat/Berth | Fare
-----
TK-RS691564-1 | Rajesh Kumar | 35 | Confirmed (CNF) | H1-1 (Lower Berth (LB)) | ₹4880.00
=====
```

Ticket Booking

10.5 PNR Status Enquiry

Shows retrieving a booking record using PNR RS412404 , displaying passenger manifest, active status, coach, and berth allocations.

```
Command Prompt - RailSync PNR Status Enquiry

Enter your choice (1-9): 4
--- PNR STATUS ENQUIRY ---
Enter PNR Number (e.g. RS861402):
===== PNR RECORD =====
PNR Number      : RS412404
Train           : 12952 - Mumbai Rajdhani
Route           : NDLS -> BCT
Date            : 2026-09-18
Class           : AC 2 Tier (2A)
Booked By       : passenger
Overall State: ACTIVE

Passenger List:
  • TK-RS412404-1 | Rahul Sharma (32 yrs) | Status: Confirmed (CNF) | Coach A1, Berth 1 (Lower Berth (LB))
  • TK-RS412404-2 | Sunita Sharma (62 yrs) | Status: Confirmed (CNF) | Coach A1, Berth 2 (Upper Berth (UB))
=====
```

PNR Enquiry

10.6 Ticket Cancellation & Refund Receipt

Shows cancelling a confirmed ticket and generating an official refund receipt with departure-based cancellation deductions and net refund payable.

```
Command Prompt - RailSync Ticket Cancellation & Refund Receipt

Enter your choice (1-9): 5
--- CANCEL TICKET ---
Enter PNR Number to cancel:
Tickets under PNR RS751217:
  1. TK-RS751217-1 - Amitabh Verma [Status: Confirmed (CNF)] Fare: ₹2050.0
  2. TK-RS751217-2 - Aarav Verma [Status: Confirmed (CNF)] Fare: ₹1055.0
Select ticket number to cancel (1-2, or 0 to abort): Are you sure you want to cancel ticket TK-RS751217-1 for Amitabh Verma? (y/n):
=====
RAILSYNC - OFFICIAL REFUND RECEIPT
=====
Receipt ID       : REF-407D3A2E
PNR Number      : RS751217
Passenger Name   : Amitabh Verma
Train Number     : 12626
Cancellation Date : 17-Sept-2026 02:27:38
=====
Original Fare Paid : ₹    2050.00
Cancellation Fee   : ₹    180.00
=====
NET REFUND PAYABLE : ₹    1870.00
=====
Status/Remarks   : Seat released back to general availability inventory.
Refund will be credited back to the original payment
method within 3-5 business days as per IR rules.
=====
```

Cancellation and Refund

10.7 RAC and Waiting List Queue Status

Displays live inspection of available confirmed seats, RAC queue capacity, and Waiting List queue capacity across all coach classes for Train 12952.

```
Command Prompt - RailSync RAC & Waiting List Queue Status

Enter your choice (1-9): 6
--- RAC & WAITING LIST QUEUE STATUS ---
Enter Train Number: Enter Journey Date (YYYY-MM-DD, press Enter for 2026-09-18):
Queue status for Train 12952 on 2026-09-18:
  • AC First Class (1A): Available Confirmed: 18 | RAC Queue: 0/ 4 | WL Queue: 0/10
  • AC 2 Tier (2A)      : Available Confirmed: 70 | RAC Queue: 0/12 | WL Queue: 0/30
  • AC 3 Tier (3A)      : Available Confirmed: 96 | RAC Queue: 0/16 | WL Queue: 0/40
```

Queue Status

10.8 Dynamic Fare & Concession Calculation

Shows distance-based fare calculation for a senior citizen passenger on Train 12952 with class multiplier and 40% statutory concession applied.


```
Command Prompt - RailSync Dynamic Fare & Concession Calculation

Enter your choice (1-9): 7
--- DYNAMIC FARE & CONCESSION CALCULATOR ---
Enter Train Number (or press Enter for default 12952): Select Class: 1) 1A 2) 2A 3) 3A 4) CC 5) SL
Choice (1-5): Enter passenger age: Enter gender (M/F/O):
Fare Calculation Breakdown:
Train      : 12952 - Mumbai Rajdhani (Rajdhani Express)
Distance   : 1384.0 km
Base Fare Rate : ₹0.95 per km
Class Selected : AC First Class (1A) (Multiplier: 3.5x)
Reservation Fee : ₹60.0
Applied Concession: General / No Concession
Total Fare   : ₹4880.00
```

Fare Calculator

10.9 Swing Graphical User Interface (Passenger Portal)

Shows the desktop Swing interface for passenger train searching and multi-passenger booking with live fare calculation.

Train Search & BookPNR Status EnquiryMy Bookings & Cancellation

From: New Delhi (NDLS)To: Mumbai Central (BCT)Journey Date (YYYY-MM-DD) 2026-09-18Class: AC First Class (1A)Search Trains

AVAILABLE TRAINS ON ROUTE

Train No	Train Name	Category	Departure	Arrival	Duration	Confirmed ...	RAC Capacity	WL Capacity	Fare (₹)
12952	Mumbai Rajdha..Rajdhani Express	16:55	08:35	15h 40m	18 Available	4	10	4880.00	

PASSENGERS TO BOOK (Max 6 per booking)

+ Add Passenger- Remove Selected

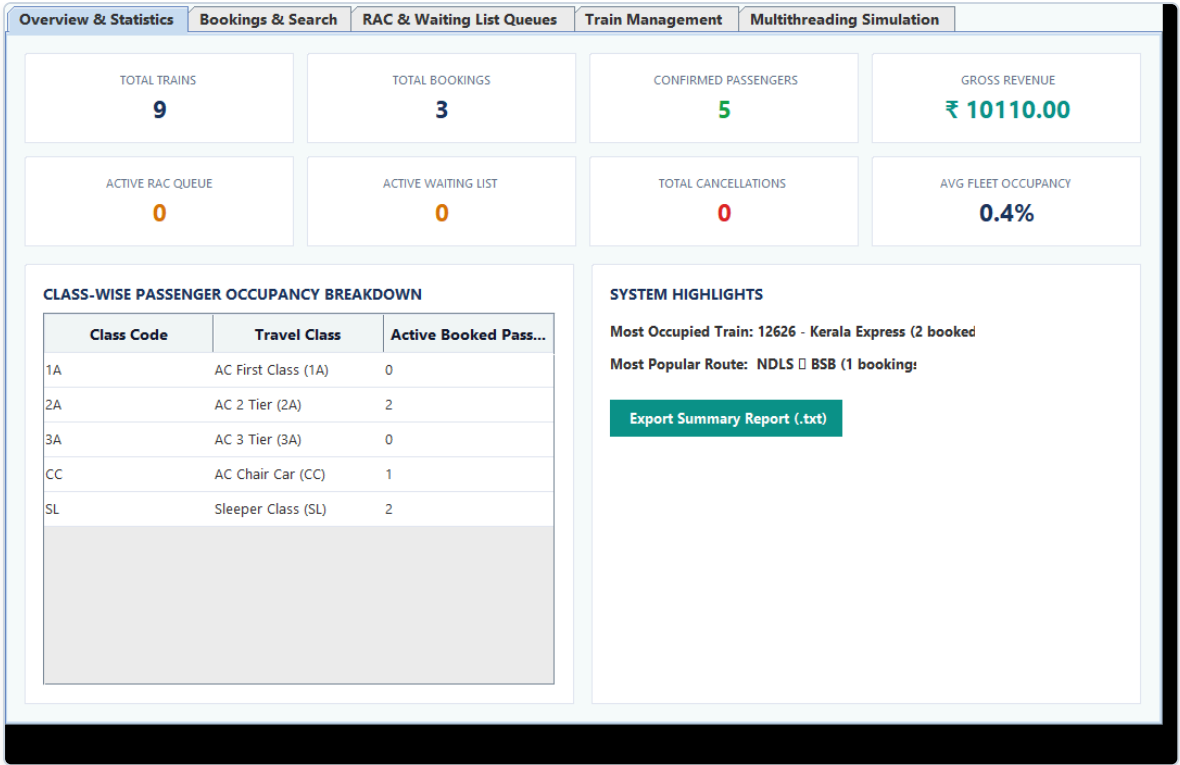
S.No	Name	Age	Gender	Phone	Berth Preference	Concession
1	Rahul Sharma	30	Male	9876543210	Lower Berth (LB)	NONE

Estimated Total Fare: ₹ 4880.00 (1 Passengers)Confirm & Book Ticket

GUI Passenger Portal

10.10 Swing Graphical User Interface (Admin Console)

Shows the administrative dashboard displaying fleet statistics, total bookings, confirmed passengers, gross revenue, and class-wise occupancy breakdown.



GUI Admin Console

10.11 Exported Electronic Reservation Slip

Shows the generated text ticket slip exported by the system with complete journey information and passenger berth roster.

```
Notepad - data/tickets/ticket_RS412404.txt (Electronic Reservation Slip)

=====
                INDIAN RAILWAYS - ELECTRONIC RESERVATION SLIP
                POWERED BY RAILSYNC SYSTEM
=====

PNR NUMBER      : RS412404          BOOKING ID      : BK10001
TRAIN NO & NAME  : 12952 Mumbai Rajdhani  TRAVEL CLASS   : AC 2 Tier (2A)
FROM STATION    : New Delhi (NDLS)      TO STATION     : Mumbai Central (BCT)
JOURNEY DATE    : 18-Sept-2026 (Fri)    BOOKED ON      : 17-Sept-2026 02:24:10
OVERALL STATUS  : ACTIVE

=====
SNO | PASSENGER NAME | AGE | GENDER | CONCESSION | STATUS / BERTH | FARE (?)
=====
1   | Rahul Sharma   | 32  | Male   | NONE       | CNF (A1-1 Lower Berth (LB)) | 3425.00
2   | Sunita Sharma  | 62  | Female | SENIOR_CITIZEN | CNF (A1-2 Upper Berth (UB)) | 2165.00
=====

TOTAL PASSENGERS : 2                TOTAL FARE PAID: ?    5590.00
=====

[BARCODE: ||| ||| | ||||| | ||||| | |||| | ||| || RS412404]
Terms & Conditions:
1. One original valid Photo ID card must be presented during journey.
2. RAC passengers are entitled to seating accommodation.
3. Waiting list passengers cannot board reserved coaches if not confirmed.
4. RailSync 24x7 Customer Support Helpline: 139
=====
```

Ticket Reservation Slip

11. Challenges Faced & Solutions

1. Race Conditions in Seat Assignment:

Problem: When multiple threads booked tickets concurrently, non-synchronized checks resulted in two passengers receiving the same berth.

Solution: Implemented fine-grained monitor locks on the individual `Train` object (`synchronized (train)`), ensuring atomicity during seat allocation while allowing concurrent bookings on different trains.

1. Managing Queue Reordering during Cancellations:

Problem: Cancelling an RAC ticket required shifting remaining queue positions without breaking FIFO order.

Solution: Utilized `java.util.LinkedList` and iterated through the queue to decrement queue positions sequentially upon removal.

1. Handling Overnight Journey Durations:

Problem: Subtracting departure time from arrival time for overnight trains resulted in negative durations.

Solution: Adjusted the duration calculation by adding 24 hours (86,400 seconds) when the arrival time is numerically earlier than departure time.

12. Conclusion & Key Takeaways

Developing RailSync provided practical experience in designing and building a modular Core Java application:

- **Object-Oriented Design:** Applied abstraction, inheritance, polymorphism, and encapsulation to create a clean, maintainable domain model.
- **Concurrency & Synchronization:** Understood how thread safety is enforced in Java using monitor locks and synchronized blocks.
- **Collections Framework:** Gained hands-on experience selecting appropriate data structures (`HashMap` , `LinkedList` , `ArrayList`) based on operational requirements.
- **GUI Programming:** Built an interactive desktop application using Java Swing with clean separation between UI components and core business logic.

13. References

1. Oracle Corporation. *The Java™ Tutorials: Learning the Java Language & Concurrency*. Oracle Documentation.
2. Oracle Corporation. *Creating a GUI With JFC/Swing*. Oracle Documentation.
3. Ministry of Railways, Government of India. *Indian Railway Standard Reservation Rules and Commercial Manual*.
4. Horstmann, C. S. *Core Java Volume I – Fundamentals*. Prentice Hall.